Explainable Machine Learning for Multi-Class Diabetes Prediction Using Human-in-the-Loop Feedback

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***Abstract*— While machine learning (ML) models achieve a high predictive accuracy in diabetes risk classification, their clinical utility is frequently constrained by "black-box" architectures, passive explanation frameworks and geographically limited datasets. This paper proposes Dia-Explain, a multi-class diabetes prediction that integrates a stacking ensemble with Explainable AI (XAI) and a Human-in-the-Loop (HitL) interactive interface. By utilizing the CDC Diabetes Health Indicators dataset, the study overcomes the demographic limitations of traditional datasets like PIMA. The proposed system establishes a causal pathway from ML prediction to clinical utility, where XAI makes the model's reasoning visible and HitL mechanisms allow users to actively explore counterfactual scenarios. Utilizing Design Science Research (DSR) methodology aligned to Peffers et al. (2007), this paper outlines the development of an artefact that shifts user interaction from passive recipient to active participant. The framework evaluates model performance using Macro F1-Score and AUC-ROC to address severe class imbalance, while prototype usability is assessed via Nielsen's heuristics. This approach lays a mathematically grounded foundation for future Agentic AI in healthcare, moving beyond static dashboards toward actionable, interactive clinical trust.**

Keywords—Machine learning, Explainable AI (XAI, Human-in-the-Loop (HitL), Diabetes Prediction, Counterfactual reasoning, Agentic AI, Design Science Research (DSR)

I. INTRODUCTION

Diabetes has affected an estimate of 589 million people globally in 2024 and projected to rise to 853 million by 2050, continuing to place a massive strain on healthcare systems [1]. Because modern health datasets are so large and complex, traditional statistical methods often fail to catch early risk indicators [2]. Machine Learning (ML) has become the standard way to solve this, offering algorithms capable of identifying non-linear risk interactions that traditional analysis overlooks [2].

Looking at recent literature, predictive modelling for diabetes has become mature. Ensemble classifiers, specifically Random Forest (RF) and XGBoost, routinely achieve accuracy scores between 80% and 90% on diabetes tasks [3], [4], [5]. To deal with the fact that these models are "black boxes", the field has adopted Explainable AI (XAI). Tools like SHAP and LIME are now widely used to generate feature attributions, effectively showing which variables pushed a prediction up or down [6], [7], [8].

The problem is that high accuracy and a static explanation chart are not always useful in a real healthcare context. Doctors still struggle to trust a system if they cannot actively question its reasoning [9], [7]. If a patient is told their blood pressure is the main risk driver, current systems generally stop there. They do not allow the user to interact with the model and ask, "What happens if I lower my blood pressure?" The explanations are post-hoc and passive.

When reviewing the literature for this study, three specific gaps stood out. First, many papers still rely on geographically limited datasets such as the PIMA Indians dataset [4]. With only 768 records from a single demographic, models trained on PIMA simply lack generalization to broader, more diverse populations. Second, while XAI generates good visualizations, very few systems actually integrate it into a Human-in-the-Loop (HitL) framework where the user can actually explore counterfactual scenarios [10], [8]. Third, the field starts looking towards Agentic AI, where systems autonomously interact with patients. There is a noticeable lack of mathematically sound, validated models underneath these agents [11]. Without a solid predictive baseline, autonomous agents risk generating unverified or "hallucinated" health interventions [12], [11].

The purpose of this study is to build a multi-class diabetes prediction system that connects complex ML modeling to actual clinical usability. Instead of just outputting a risk score, the system is designed to let users adjust their health inputs and see how their risk classification changes in real time. This approach benefits three groups. Healthcare personnel gain valuable, patient-specific explanations to support their diagnosis. Patients benefit by being able to visually explore how modifying lifestyle factors might lower their risk. Finally, the broader AI research community benefits from a reliable ML baseline that could eventually be used to safely anchor Agentic AI health assistants, rather than relying on ungrounded language models.

To achieve this, the study is guided by the following objectives:

1. To use the CDC Diabetes Health Indicators dataset to establish a multi-class baseline that overcomes the demographic limitations of PIMA.
2. To develop and evaluate a stacking ensemble model using advanced resampling techniques to handle the severe class imbalance found in the data across the three risk categories [13].
3. To integrate an explainable AI framework by using SHAP for global feature importance and LIME for local individual patient explanations [14], [8].
4. To design a Human-in-the-Loop (HitL) prototype that facilitates interactive counterfactual reasoning, explicitly architected to serve as the rigorous, mathematically

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grounded foundation required for future Agentic AI health systems.

5. To evaluate the performance and usability of the developed system using quantitative model metrics and heuristic usability assessment.

This leads to this research question: How can machine learning, combined with explainability and interactive counterfactual reasoning within a Human-in-the-Loop framework, improve transparency and clinical utility in multi-class diabetes risk prediction?

The contributions of this research are:

1) A rigorously evaluated ML pipeline using the large CDC dataset. 2) An Explainable AI framework that provides both population-level and individual patient-level reasoning and 3) A functional HitL artefact that moves beyond static dashboards. By enabling interactive counterfactual reasoning, this interface acts as the groundwork for safely integrating future Agentic AI systems into healthcare workflows.

II. Literature Review

Healthcare systems are struggling with the rising burden of diabetes and early detection methods are failing to turn data into clinical action [1]. A growing body of literature looks at ML and XAI as the solution. This review moves through four themes: Predictive Modelling, Explainable AI, Human-in-the-Loop interaction and Agentic AI. Each theme moves from the general state of the field to specific methodological choices, highlighting the gaps that justify this study's framework.

Three terms anchor this review. Machine Learning refers to algorithms that identify non-linear risk interactions in complex health datasets. Explainable AI (XAI) covers methods used to make model behavior interpretable without sacrificing predictive performance. Human-in-the-Loop (HitL) changes the user's role from a passive recipient to an active participant who can adjust inputs and see how the model reacts.

A. Predictive Modelling and the Challenge of Multi-Class Generalisation

Machine Learning has become central to clinical decision support because it can detect patterns in high-dimensional datasets that exceed human analytical capacity [2], [15]. The application of ML to diabetes prediction is heavily researched, with most studies validating supervised techniques over traditional statistics [16].

Random Forest and XGBoost are the dominant classifiers in this space. Chauhen et al. [17] established that Random Forest is resilient to overfitting, making it viable for diverse demographic groups. Kumar and Tomar [3] and Balki et al. [5] showed that XGBoost's sequential boosting strategy works exceptionally well for tabular health data. Kumar and Tomar achieved high accuracy by combing feature selection with grid-search optimization on 2000 records, while Balki et al. pushed ensemble accuracy above 85% using hyperparameter tuning. Mahajan et al. [18] evaluated multiple supervised models and found Logistic Regression achieved accuracies between 82.46% and 96% using strict binary cross-validation. Qasrawi et al. [19] validated similar approaches using glycemic control metrics to isolate specific risk factors. To tackle the severe class imbalance typical in diabetes data, Sarayu et al. [13] applied SMOTETomek to produce cleaner decision boundaries and Islam et al. [20] highlighted the value of interpretable ML in early-stage prediction using K-nearest neighbors and decision trees on PIMA data.

While Kumar, Tomar and Balki all report predictive accuracy above 85%, none of them address per-class performance when severe imbalance is present. This is a gap this study addresses directly by using SMOTETomek and macro F1 reporting. Furthermore, a critical contextual limitation persists, which is that most of these studies rely on small, homogeneous datasets, most notably the PIMA Indians Diabetes dataset. With only 768 samples from a single demographic, Shampa et al. [4] demonstrated that models trained on this data generalize poorly to wider populations. Most existing literature also focuses on binary classification (diabetic vs non-diabetic), whereas multi-class classification (no diabetes, prediabetes, diabetes) is clinically harder but far more useful. High accuracy on limited binary datasets is not enough to guarantee clinical utility. A robust predictive baseline requires large-scale, demographically diverse data such as the CDC data to ensure multi-class generalizability.

Establishing high predictive accuracy in multi-class predictions only solves half the problem. The complexity of these ensemble models creates a secondary barrier, that users cannot understand how a prediction was reached.

B. From Black-Box Models to Static Explainable AI

Relying on complex ML models has intensified concerns about the black-box problem, where end users cannot see how a model arrived at a prediction [9]. In clinical settings, this lack of transparency carries ethical and legal weight because practitioners must be able to account for their diagnostic judgements [8]. XAI has emerged as the primary technical response, offering ways to render model behavior interpretable without sacrificing performance [6].

SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) are the dominant techniques here. Mujawar et al. [7] compared these techniques on cancer patient data, confirming their utility by using Shapley values for global weighting and local surrogate models for instance-level approximations. Narrowing the focus to diabetes, Mitra et al. [14] applied SHAP for global explainability and LIME for local instance-level explanations, effectively mapping how BMI and glucose thresholds shifted individual risk profiles.

Although Mujawar and Mitra prove that XAI generates reliable feature attributions, there is a profound outcome fragmentation in how these tools are actually used. Most studies measure whether an explanation was generated (efficiency or traceability) rather than whether it improved clinical utility. Current XAI applications are post-hoc and passive. If a patient is told their blood pressure is the main risk driver, the system stops there. Static explanation charts are simply not enough to build clinical trust.

If static charts are insufficient, the next logical question is whether these explanations can be exposed to interactive user control. This is where the current literature falls notably short.

C. Limitation of Static Explanations and the Shift to HitL

Human-in-the-Loop (HitL) machine learning changes the user's role from a passive recipient to an active participant who can adjust inputs and watch the model react [10]. In chronic disease management, HitL allows patients and clinicians to influence the data shaping their predictions. When paired with XAI, the approach becomes powerful. XAI

makes the model's reasoning visible, when HitL lets users act on that reasoning through iterative input adjustment [21].

Nazar et al. [21] systematically reviewed the intersection of Human-Computer Interaction and XAI in healthcare, noting that combining visibility with action reduces cognitive load of practitioners. To facilitate this interactivity, Srushti et al. [22] demonstrated the viability of Streamlit for building interactive predictive applications, using its reactive programming framework to update predictions instantly when inputs change. Kunal et al. [23] highlighted the broader opportunities for Human and AI collaboration in future technologies.

While Nazar and Gómez-Carmona highlight the theoretical potential of combining XAI and HitL, actual implementation suffers from functional fragmentation. Very few systems integrate robust XAI into a HitL framework specifically for counterfactual reasoning in diabetes. Persistent challenges include unstable model responses to inputs outside the training distribution. Bridging the gap requires a unified framework that combines ensemble ML, dual-level XAI and an interactive HitL counterfactual interface.

This absence of validated, interactive baselines becomes critical as the field looks toward the next frontier of healthcare AI: fully autonomous systems.

D. *Agentic AI and the Need for Grounded Baselines*

The newest frontier in healthcare AI is Agentic AI, where systems autonomously interact with patients. Gridach et al. [11] surveyed the opportunities and challenges of agentic AI in healthcare, noting a rapid shift toward autonomous patient interaction.

Looking at the safety of these agents, Karunanayake [12] warns that next generation agentic AI risks generating unverified health interventions because it lacks underlying validated baselines. Gridach et al. [11] expand on this, identifying a critical missing layer beneath these agents: mathematically sound, validated predictive models.

Without a solid predictive baseline and interactive validation, autonomous agents risk generating "hallucinated" health interventions. A validated HitL framework is not just a usability feature, but a necessary, mathematically grounded stepping stone required before autonomous Agentic AI can be safely deployed in clinical workflows.

E. *Research Gap*

Looking at these four areas of literature, the same limitations keep appearing in different forms. We can group these into four specific dimensions:

First, a contextual and framing gap. Most diabetes prediction research still relies on small, geographically or demographically limited data and treats the problem as binary rather than multi-class. This is a major issue because the multi-class distinction (no diabetes, prediabetes, diabetes) is what matters most for early clinical intervention [15], [4].

Secondly is an outcome fragmentation (explanation gap). SHAP and LIME are well validated individually but researchers consistently deploy them as static, one of outputs rather than as part of an interactive system a user can actually question [14], [7].

Thirdly is a functional fragmentation (interaction gap): HitL is well theorised as a complement to XAI but very few systems actually combine the two into a working, counterfactual interface for diabetes risk specifically [11], [12].

Fourth, an inclusion and validation gap that only becomes visible once the first three are considered together. As the field moves toward agentic, autonomous healthcare systems, there is no established precedent for the kind of validated, interactive, multi-class predictive baseline those systems will eventually require [10], [21].

This study tackles all four of these gaps within a single framework, detailed in the proposed methodology below.

III. Proposed Methodology

A. *Theoretical and Conceptual Framework*

The theoretical foundation for this study is Design Science Research (DSR), following the process model outlined by Peffers et al. (2007) [24]. I chose DSR because the primary contribution of this study is a working artefact, not a hypothesis tested against existing data. This approach aligns with comparable ML healthcare research where the goal is building a practical system rather than proving a purely theoretical point [21], [20].

To show how the core constructs fit together, Figure 1 lays out the conceptual framework. A multi-class ML prediction generated by the stacking ensemble detailed in Section III.E is made interpretable through dual level Explainable AI. SHAP handles global, population level feature attribution, while LIME handles local, instance level explanations for individual predictions. But this explanation isn't the end of the road. A Human-in-the-Loop counterfactual interaction layer lets the user act on it, tweaking inputs to see how both the prediction and the explanation shift in response. Together, these three constructs prediction, explanation and interaction. Which feeds directly into the study's main outcome: clinical utility and trust. Accuracy alone isn't treated as sufficient. Looking forward, a validated system like this serves as a grounded baseline for future Agentic AI health systems, which the literature shows are currently missing this kind of interactive foundation [11].



Fig II. Conceptual framework linking ML prediction, explainability and HitL to clinical utility

B. *Research Design*

The research design explicitly follows the six activity DSR process model proposed by Peffers et al. [24]. Table 1 below maps each of the study's technical stages directly onto the corresponding DSR activity to ensure methodological rigor.

TABLE I. ALIGNMENT OF STUDY STAGES TO DSR ACTIVITES

DSR ACTIVITY (PEFFERS ET AL. 2007)	CORRESPONDING STUDY STAGE
1. Problem Identification	Established in Section II: predictive accuracy on limited, binary, demographically narrow datasets, combined with static, non-interactive explanations, limits clinical utility.
2. Definition of Objectives	The objectives stated in Section I: multi-class prediction, dual level explainability, HitL interaction and heuristic evaluation.
3. Design and Development	Stages 1-5 (Data Acquisition, Preprocessing, Stacking Ensemble, XAI Integration, HitL Prototype), detailed in Sections III.C to III.G
4. Demonstration	The artefact is exercised against representative clinical scenarios, including counterfactual queries where a user adjusts lifestyle inputs to observe changes in predicted risk.
5. Evaluation	Stage 6 (Section III.H), using quantitative model metrics (Macro F1, AUC-ROC) and heuristic usability evaluation
6. Communication	The findings and artefact are documented and disseminated through this research.

Figure 2 illustrates Stages 1 through 6 as a technical pipeline, corresponding directly to the Design, Development and Evaluation activities described above.

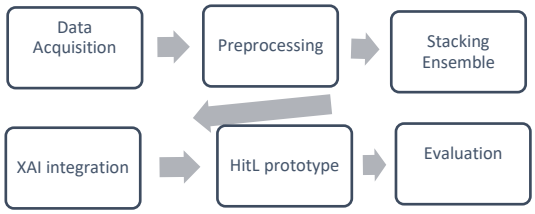


Fig III. DSR pipeline for the artefact, aligned to Peffers et al. (2007)

C. Dataset: Source and Justification

This is a quantitative study, the dataset is numerical and the model outputs classification probabilities and performance metrics drawn from clinical data [18], [5], [14]. I am using the CDC Diabetes Health Indicators dataset as the sole data source, obtained from Kaggle [25]. This is a secondary dataset meaning no primary data collection was undertaken. The data originates from the 2015 Behavioral Risk Factor Surveillance System (BRFSS). Because the underlying survey data is de-identified public health information collected by a government agency, there are no participant consent obligations for its use here.

I selected this dataset over the widely used PIMA Indians Diabetes dataset because of its scale and demographic diversity, which substantially improves model generalizability [4]. Furthermore, its three class target (no diabetes, prediabetes, diabetes) directly supports the multi-class framing argued in the literature review.

D. Dataset: Components

The dataset contains 253,680 instances across 21 numerical and categorical features, including indicators like BMI, high blood pressure status, general health rating, physical activity and smoking status. The target variable is severely imbalanced, with the majority class (no diabetes) heavily outnumbering both the prediabetes and diabetes classes. Preprocessing incorporates missing value imputation and Min-Max Scaling of continuous features. I apply SMOTETomek to the training data only to resolve this imbalance without contaminating the validation or test sets [20], [4], [13].

E. Model Development

Model development relies on a stacking ensemble where Random Forest and XGBoost act as base learners and Logistic Regression serves as the meta-model. I chose this configuration based on preliminary testing where individual classifiers underperformed specifically on minority classes. This is a common pattern reported in imbalanced diabetes classification tasks [3], [18], [5], [14].

F. Explainable AI Integration

To make the ensemble transparent, SHAP provides global feature attribution across all predictions, while LIME generates instance level explanations for individual risk classifications. Together, they address the population level and patient level explanation needs identified in the literature review.

G. Human-in-the-Loop Interactive Prototype

A Streamlit web application powers the interactive component [22]. This allows users to adjust lifestyle variables and observe how their predicted risk classification changes in response, directly turning the counterfactual interaction construct from Figure 1 into a working feature.

H. Evaluation

Performance is benchmarked using Macro F1-Score and AUC-ROC. Macro F1 is specifically chosen because it calculates the F1-score for each class independently and takes the unweighted average. This ensures the severe imbalance in the prediabetes class does not mask poor minority class performance. The prototype's usability is assessed through Nielsen's heuristic evaluation, which allows for a structured assessment of the interface without the immediate need for recruiting clinical participants [21], [16].

I. Implementation and Experimental Setup

The pipeline is built in Python. Pandas and NumPy are used for data handling, scikit-learn for the stacking ensemble and Logistic Regression meta-model and XGBoost for the boosting base learner. The imbalanced learn library handles the SMOTETomek resampling. SHAP and LIME libraries generate the explanations and Streamlit builds the interactive HitL prototype.

J. Summary

The methodology outlined above directly addresses the four gaps established in Section II.E. Using the CDC dataset with a multi-class predictive framing responds to the contextual gap, integrating SHAP and LIME within a single pipeline responds to the outcome gap, the Streamlit based counterfactual interface responds to the functional gap and building a validated, interpretable, interactive baseline responds to the inclusion gap identified in the context of emerging Agentic AI systems.

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